Apertus Swiss HVET Eval

Domain-specific factuality evaluation of Apertus v1.5 8B on Swiss Professional Education

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GitHub: `https://github.com/wortgefecht/apertus-swiss-hvet-eval` **Dataset:** `https://huggingface.co/datasets/gisiger/apertus-swiss-hvet-eval`

Summary. This project evaluates whether Apertus represents Swiss Professional Education (Higher Vocational Education and Training, HVET) accurately enough for factual information tasks in a locally specific domain. The benchmark contains 300 Swiss Standard German single-turn Q&A cases derived from 100 verified atomic facts and grounded in current SBFI and Fedlex sources. Automated SimpleQA-style grading produced 43.3% correct responses; a blinded 60-case human validation sample showed 75.0% judge-human agreement (Cohen's $\kappa = 0.515$) and indicated that the automated judge was more lenient than the human reviewer.

1 Use case and motivation

Swiss Professional Education is a useful stress test for localised factuality because it combines nationally specific qualification structures, protected or formal titles, multiple institutional responsibilities, financing rules, and current reforms. Errors are often plausible rather than nonsensical: a model may confuse a Federal Professional Examination with an Advanced Federal Professional Examination, assign recognition to the wrong authority, import terminology from Germany, or give a broadly correct answer followed by a materially false explanation.

The research question is:

How accurately does Apertus represent key concepts, structures, qualifications, institutional relationships, and selected quantitative facts in Swiss Professional Education?

The benchmark is deliberately narrow. It does not attempt to measure general reasoning, broad multilingual ability, or overall Apertus quality. Its purpose is to expose whether a general-purpose Swiss model can retrieve and apply precise local knowledge in a domain where terminology and institutional allocation matter.

2 Dataset

2.1 Construction

The benchmark was built fact-first rather than prompt-first. A curated fact bank of 100 atomic claims was created from authoritative sources with a reference date of 1 October 2026. Each fact was then represented by exactly three deliberately varied questions, yielding 300 cases. Questions are in Swiss Standard German and are single-turn, closed-book information requests.

Each evaluation record contains a stable case ID, question, canonical reference answer, accepted answer variations, and source citations. A separate metadata record stores fact ID, category, subcategory, authoring difficulty, question type, misconception target where applicable, reference date, source authority/title/URL, retrieval date, source-rights note, curation method, language, and item licence.

The final corpus uses 16 unique source URLs from two authorities: the State Secretariat for Education, Research and Innovation (SBFI/SERI) and the Swiss Confederation's Fedlex system. Each item cites on average 1.3 sources. Source-related arrays remain positionally aligned in metadata.

2.2 Coverage and quality control

The seven categories are: system structure (9), qualifications and titles (21), access/pathways/progression (30), institutions and responsibilities (21), recognition/quality/regulation (87), funding and financing (69), and 2026 reforms/edge cases (63).

Question forms include direct facts, application, misconception, responsibility, comparison, boundary, calculation, classification, relationship, date-rule, document-lookup, reverse-lookup, and several low-frequency specialist forms. The metadata field `difficulty` is an authoring taxonomy, not a psychometrically validated difficulty scale.

Quality control included exact schema validation, 100-fact/300-item count checks, source provenance checks, item-level licence checks, deterministic case ordering, normalised duplicate detection, and a near-duplicate scan. Nine high-similarity pairs remained after review; they were retained because they represent deliberate variants rather than accidental duplicate cases. The benchmark was frozen before Apertus inference.

2.3 Licensing and responsible sourcing

The benchmark and evaluation data are released under CDLA-Permissive-2.0. Code is Apache-2.0 and documentation is CC BY 4.0. External sources are cited but not redistributed or relicensed. The dataset contains no participant records, personal profiles, private conver-

sations, or sensitive personal information. Questions, reference-answer formulations, classifications, and annotations are original benchmark content.

3 Apertus inference

The final run used `swiss-ai/Apertus-v1.5-8B` at immutable revision `ef4bbc54d6fe658bc782280ff9f3b53a10a9758f`. Inference was executed with the official Swiss AI vLLM runtime through Hugging Face Jobs on an A100-large environment.

Parameter	Final value
Backend	vLLM 0.23.1rc1.dev1029+ga601a9d99
Temperature	0.0
Max completion	1024 tokens
Seed	0
Thinking	false
Max model length	4096
Batch size	50
Prompting	one user message, no system prompt
Context	closed-book, no retrieval
Run ID	`6ad08758c66288145224df76`

Table 1: Frozen Apertus inference configuration.

All 300 cases completed with `finish_reason=stop`. The longest response used 744 completion tokens, so no final response reached the 1024-token ceiling. Raw model responses were preserved unchanged.

4 Evaluation method

4.1 LLM-as-a-judge

Primary grading follows the Track 1B SimpleQA-style three-way scheme: `correct`, `incorrect`, and `not_attempted`. The final judge was `Qwen/Qwen3-235B-A22B-Instruct-2507`, served by DeepInfra via Hugging Face Inference Providers, with temperature 0 and a 1024-token output limit.

The judge prompt treats the benchmark reference answer as authoritative. A response need not repeat irrelevant reference details, but it is marked incorrect when it contradicts a necessary fact, omits a required formal entity/value, or adds a materially false claim about the concept being tested. `not_attempted` is reserved for refusals, clarification requests, or other responses without a substantive attempt.

Before the full judge run, the final rubric/model combination was checked on a deterministic 12-case calibration sample (seed 20261001). The prompt and judge configuration were then frozen. This calibration set was kept separate from subsequent human validation.

4.2 Blinded human validation

A fixed-seed random sample of 60 cases was drawn from the remaining 288 non-calibration cases (seed 20261015). Thus, 20% of the full 300-case benchmark received human review while calibration and validation remained disjoint. The human evaluator was blinded to judge labels and used the same three-way rubric.

This differs slightly from a simple random sample over all 300 judge decisions because the 12 calibration cases were excluded from the sampling frame. The choice improves independence between calibration and validation, but it is reported explicitly as a design limitation.

5 Results

Automated grading labelled 130/300 responses correct (43.3%), 165/300 incorrect (55.0%), and 5/300 not attempted (1.7%). The item-level Wilson 95% interval for the correct rate is 37.8–49.0%.

In the human-validation sample, 19/60 responses were correct (31.7%), 39/60 incorrect (65.0%), and 2/60 not attempted (3.3%). Exact judge-human agreement was 45/60 = 75.0% (Wilson 95% CI 62.8–84.2%), with Cohen's $\kappa = 0.515$.

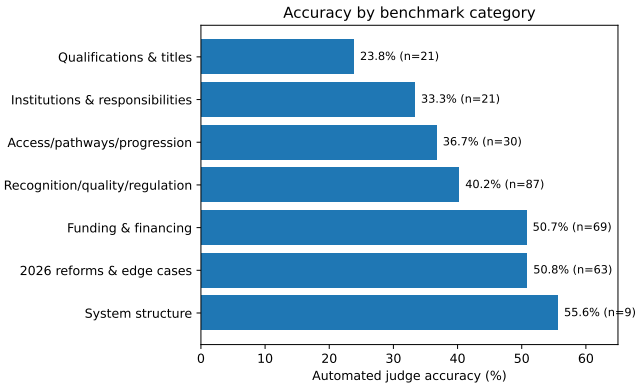


Figure 1: Automated judge accuracy by content category.

Category results show the strongest weaknesses where formal local precision matters most. Qualifications/titles scored 23.8%, institutions/responsibilities 33.3%, and access/pathways/progression 36.7%. Funding/financing and 2026 reforms/edge cases were around 51%, while the small system-structure category reached 55.6%.

The question-type breakdown is even more striking. Direct factual questions scored 19.8% and responsibility questions 8.3%, whereas misconception prompts scored 75.4%. This should not be read as a generic reasoning advantage. Many misconception items provide a contrastive frame that helps the model reject an incorrect premise, while direct-fact items demand exact Swiss terminology or values without such scaffolding.

The authoring-labelled `advanced` items scored 69.2%, compared with 42.2% for intermediate and 19.4% for basic. This counter-intuitive pattern reflects taxonomy composition, not validated difficulty: many `advanced` items are misconception or boundary cases, whereas many `basic` items are direct-fact retrieval cases.

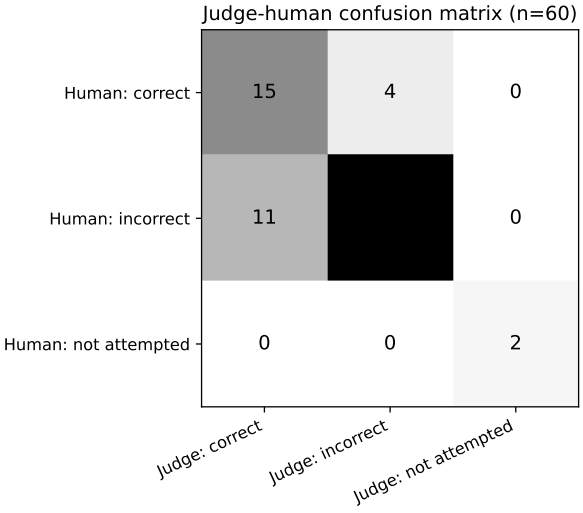


Figure 2: Judge-human confusion matrix for the 60-case validation sample.

The confusion matrix shows asymmetric disagreement. Eleven human-rated incorrect answers were labelled correct by the judge, while four human-rated correct answers were labelled incorrect. No disagreement involved `not_attempted`. The judge therefore tends to be lenient about answers that contain a correct core statement but also include a materially false domain claim. Consequently, the 43.3% automated score should not be treated as unquestioned ground truth; the 31.7% human sample estimate provides an important counterpoint.

6 Qualitative error analysis

All 15 judge-human disagreements are preserved as a dedicated analysis artefact. Recurring model-error patterns across the broader benchmark include:

- confusion between Federal Professional Examination, Advanced Federal Professional Examination, HF qualifications, and related title conventions;
- incorrect allocation of responsibilities between SBFI, examination bodies, cantons, and education providers;
- correct yes/no or high-level answers followed by materially false institutional explanations;
- importation of German or generic international education terminology into a Swiss question;
- incorrect thresholds, dates, procedural steps, or document names;
- plausible expansions of abbreviations that are factually invented.

These failures matter for the intended use case because they often produce authoritative-sounding answers that are locally wrong rather than obviously nonsensical.

7 Reproducibility

The GitHub repository contains the frozen benchmark, model responses, judge metadata, human-validation sample/results, analysis artefacts, construction and validation scripts, and exact inference metadata. Running

```
make run
```

from a clean checkout builds a Docker image and performs a deterministic offline reproduction. It validates benchmark construction, validates frozen graded Apertus responses against the recorded run metadata and input SHA-256, recomputes aggregate statistics, and verifies the headline result counts, agreement, and κ against frozen expected values. The workflow requires no API key and no GPU because it reproduces scoring from fixed model outputs. The original GPU inference and external judge require the documented hosted runtimes and credentials.

The locally tested final workflow completed successfully and reproduced 130/165/5 judge labels, 19/39/2 human labels, 45/60 agreement, and $\kappa = 0.515$ exactly.

8 Limitations and recommendations

The benchmark covers one narrow Swiss domain, one language variety, one Apertus model size, one response per item, and one human evaluator. There is no human inter-rater reliability estimate. Some question-type groups are very small and should not be interpreted independently. Because each atomic fact yields three items, observations are clustered; the reported Wilson intervals are descriptive item-level intervals rather than cluster-adjusted inferential estimates.

No dedicated contamination analysis against Apertus training data was completed. This is a relevant opportunity for a future benchmark version if a practical training-corpus search method becomes available. The benchmark is also time-sensitive: regulatory and institutional facts can change after the 1 October 2026 reference date.

For model improvement, the results suggest prioritising precise Swiss entity and title representation, institutional-role grounding, date/threshold fidelity, and mechanisms that suppress plausible but unsupported elaboration after a correct core answer. For benchmark development, a second expert rater, clustered uncertainty estimates, and controlled multilingual variants would provide the highest-value next extensions.

9 Conclusion

Apertus demonstrates partial competence in Swiss Professional Education but remains unreliable for precise domain factuality. It performs markedly better when a prompt supplies a misconception or contrastive frame than when it must retrieve exact terminology, responsibilities, or values directly. The benchmark also shows why specialised LLM evaluation should not rely on an automated judge alone: human validation exposed a systematic leniency pattern that materially affects interpretation of the headline score.

References

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